

SOFTWARE CHANGE MANAGEMENT PRACTICAL PROJECT

PROJECT SCENARIO – INITIAL VERSION

Project Name

Student Management System

Organization

ABC Institute

1. Project Background

ABC Institute currently manages student information using paper records and Excel spreadsheets. Student registration information, contact details, and other basic student records are maintained manually by administrative staff.

As the number of students increases, managing these records has become difficult. Searching for information, updating records, and maintaining accurate student data require significant time and effort.

The institute therefore wants to introduce a simple web-based **Student Management System** to provide a centralized platform for managing student records.

2. Current Problems

The current manual system has the following problems:

- Student information is stored in different files and documents.
 - Searching for student information is time-consuming.
 - Manual updates can lead to errors.
 - Duplicate or inconsistent records may be created.
 - Access to student information is difficult to control.
 - Paper records require additional storage and maintenance.
 - There is no centralized system for managing student information.
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3. Proposed Solution

ABC Institute proposes to develop a web-based **Student Management System**.

The system will provide basic authentication and student record management. Authorized administrators will be able to manage student records, while students will be able to view their own profile information.

The initial release will focus on essential functionality defined for **Version 1.0**.

Then Section 8 already explains that future features can be introduced through controlled Change Requests.

4. Initial Version – v1.0

4.1 User Authentication

The system will allow users to:

- Log in.
- Log out.
- Access functions according to their role.

4.2 Student Management

Administrators will be able to:

- Add students.
- View the student list.
- View student details.
- Update student information.
- Delete student records.

Students will be able to:

- Log in.
 - View their own profile.
 - Log out.
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5. Target Users

Administrator

The administrator is responsible for managing student records and will have access to the main student management functions.

Student

Students will use the system to securely access and view their own profile information.

6. Initial Student Information

The system will initially maintain basic student information, including:

- Student ID
- Registration Number
- First Name
- Last Name
- Email
- Phone Number

The detailed data requirements will be finalized during the Requirements Engineering phase.

7. Initial Business Rules

The following initial business rules apply:

1. Only authorized users can access the system.
2. Only administrators can create student records.
3. Only administrators can update student records.
4. Only administrators can delete student records.
5. Students can view their own profile.
6. Students cannot modify or delete student records.
7. Student registration numbers should be unique.
8. User passwords must be stored securely.
9. Student information should be maintained in a centralized database.

These rules may be refined during Requirements Engineering.

8. Initial Scope

In Scope for Version 1.0

- User authentication
- User login and logout

- Student registration
- Student record management
- Student list and details
- Student profile
- Basic role-based access
- MySQL database
- Web-based user interface
- REST API

Out of Scope for Version 1.0

- Course management
- Course enrollment
- Attendance management
- Fee management
- Timetable management
- Examination management
- Department management
- Reports
- Notifications
- Profile photo upload
- Advanced role-based access control
- Audit logs

Features outside the Version 1.0 scope may be considered in future releases through the Software Change Management process.

9. Initial Technology Environment

Component	Technology
Frontend	React
Backend	Node.js + Express
Database	MySQL
Version Control	Git
Repository	GitHub
API Testing	Postman

10. Project Objective

The primary objective is to develop a simple web-based system that allows ABC Institute to securely manage basic student information.

11. Expected Initial Outcome

The initial development cycle is expected to produce a working **Version 1.0 Student Management System** that provides:

- User authentication
- Administrator student management
- Student profile viewing
- React frontend
- Node.js/Express backend
- MySQL database
- Git/GitHub version control
- Documented requirements
- System design
- Database design
- Test documentation

After Version 1.0 is released, future requirements will be managed through controlled Change Requests.